



SMART EV CHARGING STATION CONTROLLER

Technical Project Documentation

Engineering Portfolio Edition

Platform STM32H743ZI	RTOS FreeRTOS v10
Protocol OCPP 1.6J / MQTT	Standard IEC 61851 / SAE J1772

Prepared for Engineering Portfolio, GitHub Repository & Job Applications

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1. Executive Summary

The Smart EV Charging Station Controller is a comprehensive, industry-grade embedded systems project designed to demonstrate deep technical competency across electric vehicle infrastructure, real-time operating systems, industrial communication protocols, IoT cloud integration, and power electronics. Developed on the STM32H743ZI microcontroller, a high-performance ARM Cortex-M7 device operating at up to 480 MHz. this project addresses the urgent global need for intelligent EV charging infrastructure by delivering a production-ready controller platform.

Electric vehicles are experiencing unprecedented global adoption driven by climate policy mandates, declining battery costs, and consumer demand. However, charging infrastructure particularly in emerging markets, remains critically underdeveloped. This project responds to that gap by engineering a controller that supports Level 2 AC charging (IEC 61851-1 / SAE J1772), real-time energy metering, RFID-based user authentication, cloud telemetry via MQTT, and a path toward full OCPP 1.6J compliance for interoperability with Charge Point Management Systems (CPMS).

Beyond its technical scope, this project serves as a living portfolio artifact demonstrating the full embedded systems development lifecycle: from hardware selection and schematic design to firmware architecture, protocol implementation, cloud integration, and field validation — the exact competency stack demanded by employers in EV, energy, and industrial embedded roles.

Key Project Metrics	
Microcontroller	: STM32H743ZI — ARM Cortex-M7 @ 480 MHz, 2 MB Flash, 1 MB SRAM
Charging Standard	: IEC 61851-1 Level 2 AC / SAE J1772 Pilot Signal (PWM)
RTOS	: FreeRTOS v10 — Priority-based preemptive multitasking
Energy Metering	: ATM90E36A 3-phase energy metering IC (±0.1% accuracy Class 0.1)
Communication	: MQTT over Wi-Fi/Ethernet, CAN 2.0B, Modbus RTU, OCPP 1.6J (planned)
Authentication	: RFID ISO/IEC 14443 (MFRC522), AES-128 encrypted credentials
Cloud Backend	: Firebase Realtime Database + Google Cloud Functions
Target Market	: Commercial, fleet, and public EV charging — Nigeria & Sub-Saharan Africa

2. Problem Statement

The global transition to electric mobility is accelerating. The International Energy Agency (IEA) projects over 145 million EVs on the road by 2030, with Sub-Saharan Africa representing one of the most underserved markets despite growing EV interest. Nigeria, with its population of over 220 million and a rapidly urbanising transport sector, faces structural barriers that make deploying robust EV charging infrastructure both urgent and technically challenging.

2.1 Limited Charging Infrastructure

Nigeria currently has fewer than 50 publicly accessible EV charging points, concentrated almost entirely in Lagos and Abuja. This compares to over 1.8 million public chargers in China and 900,000 across Europe. The deficit stems from a combination of grid instability, high equipment import costs, lack of local technical expertise, and absent regulatory frameworks for charge point operation — challenges that demand locally engineered, cost-optimised solutions.

2.2 Absence of Intelligent Monitoring Systems

Existing chargers in the African market, where present, are largely passive — they deliver power but collect no operational data. There is no real-time monitoring of energy delivery, no fault detection, no remote diagnostics, and no telemetry for grid operators. This opacity makes maintenance reactive rather than preventive, drives up operational costs, and undermines user confidence in charging reliability.

2.3 Energy Management and Grid Stability

Nigeria's national grid operates at approximately 4,000–6,000 MW of available generation capacity against a demand that exceeds 15,000 MW — a structural deficit. Uncontrolled EV charging presents a load aggregation risk that could cause localised grid instability. Smart charging controllers with dynamic load management, time-of-use scheduling, and demand response capabilities are essential to responsible EV adoption.

2.4 User Authentication and Revenue Management

Without secure user identification and session management, charging stations are vulnerable to energy theft, unauthorised use, and billing disputes. Implementing RFID-based authentication with encrypted session tokens and cloud-backed billing records addresses these commercial and security requirements directly.

2.5 Renewable Energy Integration

Nigeria receives over 5 kWh/m²/day of solar irradiance — among the highest in the world. Integrating solar PV and battery energy storage with EV charging stations creates distributed green energy hubs that reduce grid dependency, lower the cost of charging, and support national renewable energy targets. The controller architecture must support hybrid power source management with seamless source arbitration.

3. Project Objectives

3.1 Primary Objectives

- Design and implement a fully functional EV charging station controller on the STM32H743ZI platform.
- Implement the IEC 61851-1 / SAE J1772 pilot signal state machine for vehicle connection and charging authorisation.
- Achieve Class 0.5 or better energy metering accuracy using a dedicated metering IC.
- Deploy secure RFID-based user authentication with AES-128 encrypted session tokens.
- Implement MQTT-based cloud telemetry for real-time monitoring, remote command, and data logging.
- Develop a RTOS-based firmware architecture with clearly partitioned, deterministic tasks.
- Integrate comprehensive hardware and software fault protection aligned with IEC 61851-3 safety requirements.

3.2 Secondary Objectives

- Implement a CAN 2.0B bus interface for multi-station network communication and load coordination.
- Build an SD card data logger with circular buffer architecture for offline session storage.
- Design a modular hardware platform enabling straightforward component substitution and field upgrades.
- Develop a web-based and mobile monitoring dashboard with real-time KPIs and historical analytics.
- Establish a framework for OCPP 1.6J compliance as a future upgrade path.
- Demonstrate renewable energy source integration logic in the power management subsystem.
- Produce complete documentation suitable for open-source publication, portfolio presentation, and technical peer review.

4. System Requirements

4.1 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall authenticate users via RFID ISO/IEC 14443A cards with AES-128 encrypted session tokens.	High
FR-02	The system shall detect vehicle connection and disconnection via SAE J1772 pilot line voltage monitoring.	High
FR-03	The system shall generate a compliant PWM pilot signal (1 kHz, duty cycle 10–96%) to communicate available current capacity.	High
FR-04	The system shall measure voltage, current, active power, reactive power, apparent power, and energy (kWh) with accuracy Class 0.5.	High
FR-05	The system shall detect and respond to overcurrent, overvoltage, undervoltage, ground fault, and short-circuit conditions within 100 ms.	High
FR-06	The system shall publish telemetry data to an MQTT broker with a maximum latency of 5 seconds.	High
FR-07	The system shall support remote configuration and remote shutdown commands received via MQTT or CAN bus.	Medium
FR-08	The system shall log all charging sessions, fault events, and authentication events to SD card in structured CSV/JSON format.	High
FR-09	The system shall display charging status, session energy, elapsed time, and fault codes on a 128x64 OLED display.	Medium
FR-10	The system shall support smart scheduling: allow sessions to be pre-scheduled for off-peak energy delivery.	Medium
FR-11	The system shall interface with a mobile application via MQTT to display session status and initiate remote stop.	Low
FR-12	The system shall support OTA firmware update via MQTT-delivered binary with SHA-256 integrity verification.	Low

4.2 Non-Functional Requirements

Category	Requirement
Reliability	System shall maintain $\geq 99.5\%$ uptime. MTBF target: 50,000 hours. Watchdog timer recovery in ≤ 5 s.
Scalability	CAN bus network shall support up to 32 charging stations on a single segment. Cloud backend shall scale to 10,000 concurrent sessions.
Cybersecurity	All cloud communication encrypted with TLS 1.3. User credentials stored as bcrypt hashes. OTA binaries signed with RSA-2048.

Safety	Emergency stop shall open contactor within 20 ms. Ground fault interrupt response ≤ 50 ms. Compliant with IEC 61851-1 and IEC 62196.
Real-Time Performance	Charging state machine shall execute at 100 Hz. Pilot signal PWM jitter $\leq 1 \mu\text{s}$. CAN message latency ≤ 1 ms.
Cost Effectiveness	BOM cost target $\leq \$120$ USD for controller assembly (excluding enclosure and cabling). Suitable for local manufacturing in Nigeria.
Maintainability	All firmware modules shall carry unit tests with $\geq 80\%$ branch coverage. Hardware interfaces documented to PCB assembly level.

5. System Architecture

The system architecture is organised as a layered hierarchy of six functional layers, each with clearly defined responsibilities and inter-layer interfaces. This decomposition enables independent development, testing, and replacement of individual subsystems.

5.1 Layer 1 — User Interface Layer

The user interface layer handles all direct human-system interactions. A display, driven over I2C/SPI, presents real-time charging status, session energy accumulation, elapsed time, and fault diagnostics to the user standing at the station. A tactile keypad with four function keys provides mode selection, menu navigation, and manual session control.

RFID authentication is implemented using the MFRC522 module communicating over SPI at 10 MHz. Upon card presentation, the MCU reads the card UID, verifies it against a locally cached authorised user list (stored in internal Flash), and if necessary, validates against the cloud user registry within a 3-second timeout. Mobile application integration is provided via the cloud layer — the mobile app communicates with the station indirectly through Firebase, allowing remote session start, stop, and status monitoring.

5.2 Layer 2 — Control Layer

The STM32H743ZI Cortex-M7 MCU is the system's nerve centre. Running RTOS v10.4, it executes eight concurrent tasks with priority-based preemptive scheduling. The IEC 61851-1 state machine — the core control algorithm — runs in a dedicated high-priority task at 100 Hz, ensuring deterministic response to pilot line state transitions. Hardware timers generate the 1 kHz PWM pilot signal with $\leq 1 \mu\text{s}$ jitter, meeting SAE J1772 timing specifications.

A cascaded watchdog architecture uses both the Independent Watchdog (IWDG) and Window Watchdog (WWDG) peripherals. The IWDG provides a last-resort 5-second system reset; the WWDG monitors task scheduling integrity by requiring a service from within a defined time window.

5.3 Layer 3 — Measurement Layer

Accurate energy metering is provided by the ATM90E36A three-phase energy metering IC, connected over SPI at 4 MHz. This device provides hardware measurement of RMS voltage, RMS current, active power, reactive power, apparent power, power factor, and frequency, all with Class 0.1 accuracy (0.1% error). Calibration coefficients for voltage and current gain are stored in the MCU's internal Flash and applied at initialisation.

Current measurement uses a combination of the ATM90E36A's integrated current channel inputs and ACS758 Hall-effect current sensors on the AC power path. Voltage sensing uses precision resistive dividers with isolation transformers. A dedicated analog front end with 16-bit ADC channels on the STM32H7 provides backup measurement and overcurrent interrupt capability.

5.4 Layer 4 — Communication Layer

The communication layer implements five distinct protocol stacks operating concurrently under FreeRTOS task management. Wi-Fi connectivity is provided by the WiFi Module (ESP32-C6 for dual band), connected via UART with AT command interface — or alternatively via SPI with a custom driver for higher throughput. Ethernet is provided via the LAN8742A PHY interfacing with the STM32H7's integrated MAC, enabling wired network operation in commercial installations.

CAN bus communication uses the TCAN332 transceiver connected to the STM32H7's bxCAN peripheral, supporting CAN 2.0B at 500 kbps for multi-station coordination. UART ports serve Modbus RTU communication with external energy sub-meters and building management systems. SPI serves the RFID reader, energy metering IC, and SD card simultaneously, multiplexed via chip-select GPIO lines.

5.5 Layer 5 — Cloud Layer

The cloud backend is implemented on Google Firebase, leveraging the Realtime Database for live session data and Firestore for historical records and user management. Firebase Cloud Functions implement business logic: tariff calculation, session validation, alert generation, and Firestore write triggers. A web-based admin dashboard built in React.js consumes data via Firebase SDK, presenting real-time KPIs including active sessions, total energy delivered, revenue tracking, and fault history. A Progressive Web App (PWA) serves as the mobile client.

5.6 Layer 6 — Power Layer

The power layer encompasses the complete AC power handling chain from input mains to the charging output. A 32A Type-B residual current device (RCD) provides ground fault and differential current protection at the AC input. A solid-state or electromechanical contactor (rated 40A, 240V AC) under MCU GPIO control switches the charging output. The MCU's contactor driver uses a flyback-suppressed MOSFET gate driver with opto-isolation to protect the control circuitry from power-domain transients. A 24V SMPS supplies the controller board, with a 3.3V LDO regulator powering the MCU and digital peripherals.

6. Hardware Design

6.1 Component Selection

Component	Selection	Rationale
Microcontroller	STM32H743ZI (LQFP-144)	ARM Cortex-M7 @480 MHz, 2 MB Flash, 1 MB SRAM, dual-core capable, hardware FPU, 16-bit ADC, 3x SPI, 4x UART, bxCAN, Ethernet MAC, TrustZone.
Energy Meter IC	ATM90E36A (QFN-48)	Class 0.1 accuracy 3-phase metering, SPI interface, integrated PGA, UART waveform output, active/reactive/apparent energy accumulation, power quality monitoring.
Current Sensor	ACS758LCB-100B (Hall Effect)	100A range, $\pm 0.5\%$ linearity, 3 kVRMS isolation, ratiometric output compatible with STM32H7 ADC. Bidirectional — supports V2G readiness.
RFID Module	MFRC522 (SPI)	ISO/IEC 14443A/B, Mifare support, 10 MHz SPI, 13.56 MHz ISM band. Low-cost, well-documented, extensive library support.
Display	1024 x 600 HMI display	Ultra-low power, wide viewing angle, high contrast, suitable for outdoor readability with hood.
Wi-Fi Module	ESP32-C6 (UART/SPI)	Wi-Fi 6 (802.11ax), Bluetooth LE 5.0, RISC-V core, AT firmware + custom driver options, 20 dBm TX power for extended range.
Ethernet PHY	LAN8742A (RMII)	STM32H7 native RMII interface support, auto-negotiation, 10/100 Mbps, HP Auto-MDIX, low power standby, LQFP-32.
CAN Transceiver	TCAN332 (SOP-8)	3.3V compatible, 1 Mbps, ISO 11898-2, split-termination support, thermal shutdown, -40 to +125°C operation.
SD Card	SPI SD card socket + FatFS	Supports 32 GB SDHC, 4-wire SPI mode, FatFS middleware on STM32H7, DMA-assisted write for non-blocking data logging.
RTC Module	DS3231 (I2C)	Temperature-compensated crystal oscillator (TCXO), ± 2 ppm accuracy, battery-backed SRAM, 3.3V compatible, alarm outputs for scheduled charging.
Relay/Contactor	Schneider LC1D25 + MOC3023	25A 240VAC contactor, optoisolated triac driver, zero-cross switching to reduce electromagnetic interference.
Power Supply	Mean Well LRS-100-24 + AMS1117-3.3	DIN-rail 24V/100W SMPS for 97% efficiency, UL/CE certified; AMS1117-3.3 LDO for analog domain power isolation.
Emergency Stop	NC Mushroom Head PB + hardware latch	Normally-closed pushbutton in series with contactor coil circuit; hardware-latched open by

		MCU via relay — cannot be re-enabled by firmware alone.
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6.2 Pilot Signal Circuit

The IEC 61851-1 Control Pilot (CP) circuit is the most safety-critical analog subsystem. The pilot signal is a $\pm 12\text{V}$ square wave at 1 kHz with duty cycle encoding available current capacity (10% = 6A, 16% = 10A, 25% = 16A, up to 96% = 80A). The STM32H7 generates a 3.3V PWM signal that is level-shifted to $\pm 12\text{V}$ using a rail-to-rail op-amp (TL071) powered from a dual $\pm 12\text{V}$ supply.

The vehicle presents a known resistor network ($R1 = 2.74\text{ k}\Omega$ for State B, $R1 + R2 = 882\text{ }\Omega$ for State C) between the pilot line and protective earth. The controller monitors the resulting pilot voltage using the STM32H7 ADC with a precision voltage divider and input protection Zener diode. ADC readings are mapped to pilot states in firmware with hysteresis to prevent rapid oscillation on noisy lines.

7. Software Architecture

7.1 FreeRTOS Task Architecture

The firmware is structured as eight FreeRTOS tasks, each with defined stack size, priority, and inter-task communication channels. The following table documents the complete task manifest:

Task Name	Priority / Period	Responsibility
vChargingStateMachine	High / 10 ms (100 Hz)	IEC 61851-1 pilot state monitoring, contactor control, session gating. This is the safety-critical core — highest priority.
vEnergyMeter	High / 100 ms (10 Hz)	Polls ATM90E36A via SPI, updates shared energy accumulator, triggers overcurrent interrupt handling.
vMQTTPublish	Medium / 5 s	Serialises session snapshot to JSON and publishes to MQTT broker over Wi-Fi or Ethernet. Runs from a ring-buffer queue.
vDataLogger	Medium / 60 s	Writes session records to SD card via FatFS. Uses DMA for non-blocking writes. Manages log rotation.
vRFIDAuth	Medium / Event-driven	Monitors MFRC522 interrupt, reads card UID, performs local lookup, triggers cloud validation if needed.
vDisplayUpdate	Low / 250 ms (4 Hz)	Renders OLED screen: current state, session kWh, elapsed time, fault codes. Double-buffered to prevent tearing.
vCANComms	Medium / 50 ms	Transmits station telemetry on CAN bus, receives load-limit commands from coordinator node, updates duty-cycle setpoint.
vHeartbeat	Low / 1 s	Toggles status LED, services Window Watchdog, aggregates system health metrics for MQTT heartbeat message.

7.2 Inter-Task Communication

FreeRTOS queues and semaphores are used for all inter-task data exchange. Direct global variable access is prohibited in the firmware design. The following shared objects are defined:

- xEnergyQueue (Queue, depth 10): ATM90E36A readings from vEnergyMeter to vMQTTPublish and vDataLogger.
- xAuthQueue (Queue, depth 5): RFID authentication results from vRFIDAuth to vChargingStateMachine.
- xChargingMutex (Mutex): Protects the ChargingSession_t shared structure accessed by multiple tasks.
- xContactor Semaphore (Binary): Used by fault handlers and emergency stop ISR to signal contactor open request.
- xCANRxQueue (Queue, depth 20): CAN frame buffer from CAN RX interrupt to vCANComms task.

7.3 Interrupt Handling

Hardware interrupts are used sparingly and only for time-critical events. ISRs are kept short — they post to FreeRTOS queues or set task notification flags, deferring processing to the appropriate task. Critical ISRs include:

- TIM1 Update IRQ: Generates 1 kHz PWM pilot signal — no RTOS calls, direct hardware register access only.
- EXTI Line 5 (RFID): Rising edge interrupt on MFRC522 IRQ pin — posts xRFIDEvent notification to vRFIDAuth.
- ADC1 Analog Watchdog: Fires on overcurrent threshold breach — directly posts to xContactor semaphore and sets fault flag.
- CAN1 RX0 IRQ: Reads CAN FIFO0, posts frame to xCANRxQueue, immediately exits.
- IWDG timeout: Hardware reset — logged to backup SRAM for post-mortem analysis.

7.4 State Machine Design

The IEC 61851-1 state machine is implemented as a table-driven state machine with explicit transition guards and entry/exit actions for each state. Transitions are conditioned on pilot voltage ADC readings, authentication status, fault flags, and commanded state from MQTT/CAN. The machine is evaluated at 100 Hz.

8. Charging State Machine

The charging state machine is the primary safety-critical control algorithm. It implements the IEC 61851-1 standard for EV supply equipment (EVSE) control pilot behaviour, with extensions for RFID authentication gating and remote command processing.

State	Pilot Voltage / Condition	System Action
IDLE (A)	+12V DC (no PWM) — No vehicle	Contactor OPEN. Display shows 'Ready'. RFID reader active. MQTT heartbeat running.
AUTHENTICATION	Card presented — Pilot still +12V	RFID card read initiated. Cloud/local lookup. Display: 'Authenticating...'. 10 s timeout returns to IDLE if no valid card.
VEHICLE_CONNECTED (B)	+9V on pilot (vehicle R1 = 2.74 k Ω)	Authentication confirmed. PWM pilot signal activated at authorised duty cycle. Display: 'Vehicle Connected'. Waiting for vehicle to signal ready.
CHARGING_READY	PWM pilot generated, vehicle not yet drawing	Contactor OPEN. 30 s timeout if vehicle fails to transition. Monitors pilot for State C.
CHARGING_ACTIVE (C)	+6V on pilot (vehicle R1+R2 = 882 Ω)	Contactor CLOSES. Energy metering starts. Session timer starts. MQTT session-start event published. Real-time kWh accumulated.
CHARGING_COMPLETE	User stops session / SoC target reached	Contactor OPENS. Final session record written to SD card and MQTT. Display: 'Session Complete — X.XX kWh'. Return to IDLE after 30 s.
FAULT	Any fault flag asserted	Contactor OPENS immediately. Fault code displayed and published. Audible buzzer alert. Requires manual acknowledgement or remote reset to clear.
EMERGENCY_STOP (E)	Hardware E-STOP button pressed	Contactor OPENS via hardware path (independent of firmware). MCU latches fault state. Remote reset required from cloud/CAN. All sessions invalidated.

8.1 Transition Guards

All state transitions incorporate guard conditions to prevent erroneous transitions under noise or transient conditions. Pilot voltage readings require 3 consecutive samples within the target window ($\pm 0.5V$) before a state transition is accepted. This 30 ms debounce window filters transients induced by relay switching and cable inductance.

8.2 Emergency Stop Hardware Path

The emergency stop circuit operates entirely in hardware, independent of firmware. The E-STOP button (normally-closed) is wired in series with the contactor control relay coil. Pressing the button immediately de-energises the contactor coil regardless of MCU state. The MCU detects the button press via a dedicated GPIO interrupt and enters

the EMERGENCY_STOP state, logging the event and preventing the contactor from being re-energised until a remote reset command is received.

9. Communication Protocols

9.1 MQTT — Cloud Telemetry

MQTT v3.1.1 is implemented using the Eclipse Paho MQTT C Client library, ported to FreeRTOS. The controller connects to a cloud MQTT broker (EMQX or AWS IoT Core) using TLS 1.3 over TCP/IP. The connection lifecycle — TCP socket, TLS handshake, MQTT CONNECT, SUBSCRIBE — is managed by vMQTTPublish with exponential backoff reconnection.

Topic	Direction	Payload
evcs/{stationId}/telemetry	Publish (5 s)	JSON: voltage, current, power, energy_kwh, state, session_id, timestamp
evcs/{stationId}/session/start	Publish (event)	JSON: session_id, user_id, start_time, auth_method
evcs/{stationId}/session/end	Publish (event)	JSON: session_id, end_time, energy_kwh, duration_s, stop_reason
evcs/{stationId}/fault	Publish (event)	JSON: fault_code, fault_description, timestamp, recommended_action
evcs/{stationId}/cmd	Subscribe	JSON: command (start/stop/reset/setDutyCycle), parameters
evcs/{stationId}/ota	Subscribe	JSON: firmware_url, sha256_hash, version

9.2 CAN Bus — Station Network

CAN 2.0B at 500 kbps is used for multi-station coordination on a shared charging site. A designated Master Node (which may be a Raspberry Pi running a site controller) broadcasts load-limit commands; each station responds with telemetry frames. Message IDs are allocated in a priority hierarchy — fault messages hold the lowest CAN ID (highest priority).

9.3 Modbus RTU — External Meters

UART-based Modbus RTU at 9600 baud, 8N1 is implemented for communication with external DIN-rail energy sub-meters and building management systems. The STM32H7 acts as Modbus Master. Register map follows the standard Modbus energy meter register layout (0x0000–0x001B for energy registers).

9.4 OCPP 1.6J Readiness

Open Charge Point Protocol (OCPP) 1.6J over WebSockets is planned as a Phase 2 firmware upgrade. The state machine, message log architecture, and session data schema are designed with OCPP compatibility in mind. Core OCPP message types that the architecture pre-supports include: BootNotification, StatusNotification, StartTransaction, StopTransaction, MeterValues, Heartbeat, and RemoteStartTransaction.

10. Safety Features

Safety is a non-negotiable design priority. The controller implements a defence-in-depth safety architecture with independent hardware and software protection layers, ensuring that no single fault — hardware or firmware — can result in an unsafe condition for the user or the electrical installation.

Protection	Implementation
Overcurrent Protection (Software)	Energy metering IC raises an interrupt when RMS current exceeds 105% of the configured limit for >100 ms. ISR posts to xContactor semaphore. Contactor opens in ≤ 20 ms from interrupt.
Overcurrent Protection (Hardware)	AC MCB (miniature circuit breaker) rated at 125% of maximum charging current provides independent overcurrent trip. Not firmware-controllable.
Overvoltage Protection	ADC Analog Watchdog monitors pilot-line derived voltage. Trip threshold: 265V AC RMS. Contactor opens; fault state entered. TVS diode array protects electronics from transients.
Undervoltage Protection	System monitors mains voltage via ATM90E36A. Below 180V AC, charging is suspended with UNDERVOLTAGE fault code. Hysteresis: resumes above 200V.
Ground Fault Detection	Type-B 30mA RCD at AC input provides hardware GFCI protection. MCU monitors RCD trip output via GPIO; enters FAULT state on trip. Covers DC fault currents (Type-B covers AC, pulsating DC, and smooth DC).
Emergency Stop	Hardware NC pushbutton wired in series with contactor coil — firmware-independent. Audible buzzer on E-STOP. Remote reset required from cloud portal.
Thermal Protection	NTC thermistor monitors contactor, SMPS, and PCB temperature. Software trip at 75°C (warning), 90°C (charging suspended). Independent thermostat relay at 95°C.
Short-Circuit Protection	AC MCB and fast-blow fuse (125A) provide sub-cycle short-circuit interruption. Fuse rupture detection via GPIO-monitored continuity check.
Pilot Signal Integrity	Pilot signal is monitored by a separate ADC channel. If PWM output does not match ADC readback within $\pm 5\%$, a pilot-fault is declared and charging is suspended.
Watchdog Recovery	IWDG provides 5-second hardware reset. WWDG ensures no task is blocked longer than 500 ms. All resets logged to backup SRAM for diagnostic retrieval.

11. Smart Features

11.1 Dynamic Load Balancing

In a multi-station installation, dynamic load balancing prevents aggregate charging load from exceeding the site's grid connection capacity. The site controller (Master CAN Node) continuously monitors the sum of active charging currents across all stations and broadcasts load-limit commands — adjusting the pilot PWM duty cycle of individual stations — to keep total load within the contracted grid capacity. This is implemented using a simple proportional allocation algorithm with configurable priority classes for premium users.

11.2 Remote Monitoring and Management

The Firebase Realtime Database provides sub-second data propagation to the admin dashboard. Operations staff can view all active sessions, energy totals, fault states, and network connectivity status across all deployed stations from a single browser interface. Remote commands — start, stop, reset, change charging current limit — are delivered via MQTT and acknowledged back to the dashboard.

11.3 Energy Consumption Analytics

Session energy data is aggregated in Firestore and processed by scheduled Cloud Functions to produce daily, weekly, and monthly energy reports per station and per user. Analytics include total kWh delivered, peak demand periods, carbon offset estimates, revenue per station, and utilisation rate. These reports are accessible via the admin dashboard and exportable as CSV.

11.4 Smart Scheduling

Users can pre-schedule charging sessions via the mobile app to target off-peak tariff periods. The scheduling engine, running in Firebase Cloud Functions, sends a MQTT start command to the station at the scheduled time. The station's DS3231 RTC also stores a local schedule as a fallback for when cloud connectivity is unavailable.

11.5 Predictive Maintenance

Firmware tracks contactor cycle count, total energy delivered, and cumulative on-time. Cloud analytics flag maintenance alerts when these values approach manufacturer-specified service intervals. Fault frequency trending (repeated overcurrent trips, RCD trips) automatically generates maintenance work orders in the admin system.

11.6 Renewable Energy Integration

A dedicated power management mode supports hybrid solar-grid operation. When a solar inverter with Modbus interface reports surplus power generation, the controller automatically raises the charging current setpoint (pilot PWM duty cycle) to preferentially absorb available renewable energy. During grid-only periods, current is reduced to the contracted minimum. This 'solar boost' logic significantly reduces the net cost of charging for operators with on-site PV.

12. Database Design

12.1 Firebase Realtime Database Structure

The Firebase Realtime Database stores live operational data requiring sub-second update propagation. Historical data is archived to Firestore by scheduled Cloud Functions.

```
{
  "stations": {
    "EVCS-NG-001": {
      "status": "CHARGING_ACTIVE",
      "connected": true,
      "lastSeen": "2026-06-18T14:32:10Z",
      "firmware": "v1.2.3",
      "location": { "lat": 6.5244, "lng": 3.3792, "site": "Lekki Phase 1" },
      "telemetry": {
        "voltage_V": 231.4,
        "current_A": 24.8,
        "power_kW": 5.73,
        "energy_kWh": 12.45,
        "frequency_Hz": 49.97,
        "powerFactor": 0.998,
        "pilotState": "C",
        "temperature_C": 42.1
      },
    },
    "activeSession": {
      "sessionId": "sess_20260618_143000_001",
      "userId": "usr_abc123",
      "startTime": "2026-06-18T14:30:00Z",
      "energyDelivered_kWh": 12.45,
      "durationSeconds": 7330,
      "authMethod": "RFID"
    }
  },
  "faults": {
    "EVCS-NG-001": {
      "active": false,
      "lastFault": {
        "code": "F03",
        "description": "Overcurrent — 26.1A exceeded 25A limit",
        "timestamp": "2026-06-17T09:12:45Z",
        "resolved": true
      }
    }
  }
}
```

```
}  
}
```

12.2 Firestore Collections

Collection	Purpose & Key Fields
users	userId, name, email, rfidUID (hashed), authLevel, createdAt, totalSessions, totalEnergyKWh, balance
sessions	sessionId, stationId, userId, startTime, endTime, energyKWh, durationSeconds, stopReason, tariffApplied, cost
faultLogs	faultId, stationId, code, description, timestamp, resolvedAt, resolvedBy, technicalDetails
devices	stationId, firmwareVersion, hardwareRevision, installDate, location, config (JSON), maintenanceLog
otaUpdates	updateId, version, releaseDate, sha256, targetStations, rolloutPercentage, status
tariffs	tariffId, name, peakRate, offPeakRate, schedule (JSON), applicableSites

13. Dashboard Design

13.1 Admin Operations Dashboard

The admin dashboard, built with React.js and Firebase SDK, provides a comprehensive operational view accessible from any web browser. The primary view displays a real-time network map overlaid with station status icons — green (charging active), blue (idle), orange (fault), grey (offline). A sidebar panel shows aggregate KPIs: total active sessions, total power draw (kW), today's energy delivery (kWh), and total revenue (NGN).

Individual station drill-down shows live telemetry charts (voltage, current, power over time), session history, fault log, and remote command controls. Alert notifications are delivered via browser push notifications and email for critical faults (overcurrent, ground fault, offline >5 minutes).

13.2 User Mobile Dashboard (PWA)

The user-facing Progressive Web App allows registered users to view their session history, current session status, and energy consumption statistics. Users can initiate a remote start (if their RFID card is pre-registered), view their charging cost, and download session receipts as PDF. The app also displays nearest available charging stations on a map using the Google Maps API.

13.3 Analytics Dashboard

A dedicated analytics view — built with Recharts — presents 30/90-day trends for energy delivery per station, utilisation rates (percentage of time a station is actively charging), user growth, revenue trends, and fault frequency distributions. This view targets operations managers and site owners making investment decisions.

14. Cybersecurity Considerations

14.1 Authentication

All cloud API access requires Firebase Authentication tokens (JWT, RS256 signed). Admin dashboard requires multi-factor authentication (MFA). Station-to-broker MQTT connections use X.509 client certificates issued per device. RFID card UIDs are never stored in plaintext — they are salted and hashed (SHA-256) before comparison.

14.2 Secure Communication

All MQTT communication is transported over TLS 1.3 with mutual authentication (station presents client certificate; broker presents server certificate). Certificate pinning in firmware prevents man-in-the-middle attacks even if a CA is compromised. Ethernet communication uses the same TLS stack via LwIP's mbedTLS integration.

14.3 OTA Firmware Update Security

OTA update images are signed with RSA-2048 during the build pipeline (GitHub Actions CI). The firmware verifies the signature and SHA-256 hash of every received binary before writing to the secondary Flash bank. STM32H7's dual-bank Flash architecture enables atomic OTA: the new firmware is written to Bank 2 while Bank 1 continues running; swap occurs only after successful verification. A failed OTA automatically reverts to the previous firmware.

14.4 Data Protection

Personal user data (names, email, RFID UIDs) is stored in Firestore with field-level security rules. Firebase Security Rules prevent any unauthenticated access and restrict user data access to the data owner and admin roles. Session data is retained for 24 months per regulatory requirement; older records are archived to Google Cloud Storage in encrypted form.

15. Testing and Validation Plan

Test Phase	Test Cases & Pass Criteria
Unit Testing (Embedded)	State machine transition logic tested with Unity framework against all 64 valid state/event combinations. ADC reading conversion functions verified against known reference inputs. Energy accumulation algorithm validated against ATM90E36A datasheet test vectors. Target: 100% state transition coverage, $\geq 80\%$ line coverage.
Integration Testing	Full hardware-in-the-loop (HIL) bench with EV simulator (resistive network mimicking SAE J1772 states). Verify contactor response time < 20 ms from fault detection. Verify correct pilot signal generation across full duty cycle range. MQTT publish latency measured over 24-hour soak test.
Communication Testing	MQTT publish/subscribe reliability over 72-hour test with deliberate network interruptions — verify exponential backoff reconnection and no message loss. CAN bus load test at 80% bus utilisation. Modbus register read/write correctness verification against reference meter.
Safety and Fault Testing	Inject each fault condition (overcurrent, overvoltage, undervoltage, ground fault, temperature) and verify: correct state transition to FAULT, contactor opening, correct fault code displayed and published, correct recovery behavior. E-STOP hardware path verified with oscilloscope — contactor open time < 10 ms.
Load and Stress Testing	72-hour continuous charging session at 25A — monitor contactor temperature, PCB temperature, energy accumulation accuracy drift. MQTT stress test: 100 concurrent stations simulated via load test tool.
Field Validation	Pilot deployment at a controlled site (office building or residential complex) — 30-day operational monitoring. Track fault rates, MQTT uptime, energy meter accuracy (verified against calibrated reference meter), user authentication success rate.
OTA Update Testing	Deploy OTA update to 5 test units simultaneously. Verify: signature verification reject of corrupted image, successful apply of valid image, bank swap and reboot, rollback on power-cycle during write, version reporting after successful update.

16. Future Improvements

16.1 Full OCPP 1.6J Implementation

Phase 2 development will implement the complete OCPP 1.6J WebSocket-based protocol stack using the MicroOCPP library (already validated in the EVSE learning project). This will enable the station to connect to commercial Charge Point Management Systems (CPMS) — including open-source options like SteVe and CitrineOS — and participate in demand response programs managed by distribution system operators.

16.2 Vehicle-to-Grid (V2G) Support

V2G capability requires a bidirectional AC/DC power converter and implementation of ISO 15118-2 (DC Coupled) or ISO 15118-20 protocols. The current hardware platform provides the bidirectional current sensing (ACS758) and communication infrastructure required. The power stage would be upgraded to a 3.3 kW bidirectional on-board charger (OBC) unit. V2G is commercially deployed in the Netherlands and UK; regulatory frameworks in Nigeria are nascent but developing.

16.3 Solar and Battery Storage Integration

Integration with a solar MPPT controller (via Modbus) and a battery energy storage system (BESS, via CAN) would enable fully off-grid or grid-edge operation — critical for Nigeria's energy context. The 'solar boost' feature in Section 11.6 is the first step; a full energy management system (EMS) with optimisation logic based on solar forecast data, grid tariff schedules, and EV departure time preferences would follow.

16.4 AI-Based Charging Optimisation

A machine learning model trained on historical session data can predict charging demand at each station by time-of-day, day-of-week, and weather. This prediction can be used to pre-position renewable energy storage, pre-negotiate grid capacity contracts, and optimise tariff structures. TensorFlow Lite inference could run on the STM32H7's DSP-accelerated Cortex-M7 for on-device demand forecasting without cloud round-trip latency.

17. Project Timeline

Phase	Duration	Key Deliverables
Phase 1: Hardware Design	Weeks 1–4	Schematic capture (KiCad), BOM finalisation, PCB layout review, component procurement, breadboard prototype of pilot signal circuit and energy meter interface.
Phase 2: Embedded Firmware — Core	Weeks 5–10	FreeRTOS task skeleton, IEC 61851-1 state machine, pilot signal PWM, energy metering driver (ATM90E36A SPI), RFID driver, OLED display driver, SD card logging (FatFS).
Phase 3: Embedded Firmware — Communications	Weeks 11–14	MQTT stack integration (Paho), CAN bus driver, Modbus RTU master, OTA update mechanism, safety fault handling, watchdog architecture.
Phase 4: Cloud Integration	Weeks 15–18	Firebase Realtime Database schema deployment, Cloud Functions for session processing and alerts, MQTT broker configuration (EMQX), device-to-cloud data pipeline validation.
Phase 5: Dashboard Development	Weeks 19–22	React.js admin dashboard, Firebase SDK integration, real-time telemetry charts (Recharts), PWA mobile user app, analytics dashboard.
Phase 6: Testing & Validation	Weeks 23–26	Unit test suite (Unity), HIL integration testing, 72-hour soak test, safety fault injection, field pilot deployment, documentation and GitHub publication.

18. Expected Learning Outcomes

Domain	Skills Acquired
STM32 / ARM Embedded Systems	STM32H7 peripheral configuration (TIM, SPI, I2C, UART, CAN, ADC, DAC, DMA, Ethernet MAC). HAL and LL driver usage. CubeMX code generation. Flash programming and debug via SWD/JTAG.
FreeRTOS	Task creation, priority design, stack sizing. Queue, mutex, semaphore, event group usage. Watchdog integration. Deadlock prevention. Memory management (heap_4). Real-time debugging with FreeRTOS-aware JTAG tools.
Power Electronics & EV Standards	IEC 61851-1 pilot signal design and implementation. AC contactor specification and drive circuits. Optoisolation design. EMC filtering. RCD selection. Ground fault detection.
Energy Metering	3-phase energy metering IC integration. Calibration methodology. RMS calculation theory. Power quality measurement. Class accuracy standards.
IoT & Cloud Architecture	MQTT protocol stack, TLS implementation, broker configuration. Firebase Realtime Database and Firestore data modelling. Cloud Functions serverless backend. REST API design.
Industrial Communications	CAN 2.0B frame structure, arbitration, error handling. Modbus RTU register map design and master implementation. OCPP protocol architecture and WebSocket transport.
Cybersecurity	TLS mutual authentication, X.509 certificates. RSA-2048 firmware signing. OTA update security architecture. Data encryption at rest and in transit.
Software Engineering	Test-driven development with Unity framework. CMake build system. Git version control and CI/CD pipeline (GitHub Actions). Technical documentation to engineering standard.

19. Resume Impact

This project is explicitly engineered to demonstrate the competency profile sought by employers in five high-growth technical domains. The following table maps project deliverables to specific job description requirements encountered in embedded systems, EV, and energy engineering roles.

Target Role	Demonstrated Competency
Embedded Systems Engineer	STM32 firmware development, FreeRTOS RTOS, peripheral driver implementation, interrupt architecture, watchdog design, DMA, unit testing with Unity. Full C firmware codebase with CMake build system on GitHub.
EV / EVSE Engineer	IEC 61851-1 state machine implementation, SAE J1772 pilot signal design, OCPP 1.6J protocol familiarity, contactor control, ground fault detection. Demonstrates knowledge of EV charging standards at implementation level — rare for graduate candidates.
IoT / Connected Systems Engineer	End-to-end IoT pipeline: sensor → MCU → MQTT → Firebase → dashboard. TLS security implementation, OTA firmware update architecture, multi-protocol communication stack. Firebase, Cloud Functions, React.js.
Energy Systems / Smart Grid Engineer	Energy metering IC integration (Class 0.1), dynamic load balancing algorithm, solar PV integration logic, demand response architecture, time-of-use scheduling. Demonstrates intersection of embedded systems and energy systems.
Industrial Automation Engineer	CAN bus network design, Modbus RTU implementation, multi-node communication architecture, industrial-grade fault protection (overcurrent, ground fault, thermal). Directly applicable to PLC/industrial controller environments.
Renewable Energy Engineer	Solar boost charging logic, BESS integration architecture, energy management system design, off-grid operation framework. Relevant to solar EV charging deployment projects across Sub-Saharan Africa.

Portfolio Presentation Points:

- GitHub repository with complete firmware source, hardware schematics (KiCad), BOM, and documentation.
- LinkedIn project feature post with demo video of live charging session, OLED display, and real-time Firebase dashboard.
- IEEE student paper submission: 'Design of a Low-Cost Smart EV Charging Controller for Sub-Saharan African Infrastructure'.
- Technical competition entry: IEEE PES Student Poster Session / NSBE Technical Competition.

20. Conclusion

The Smart EV Charging Station Controller project represents a technically ambitious, contextually relevant, and professionally executed engineering portfolio work. It addresses a genuine infrastructure deficit facing Nigeria and Sub-Saharan Africa by delivering a system that is not merely functional but production-aligned — incorporating the safety standards, communication protocols, security architecture, and cloud integration that commercial EV charging deployments require.

Technically, the project demonstrates mastery of the complete embedded systems development stack: from low-level peripheral driver implementation on the STM32H743ZI to real-time task scheduling in FreeRTOS, from industrial communication protocols (CAN, Modbus, OCPP) to secure cloud telemetry and a live operational dashboard. The IEC 61851-1 state machine implementation, SAE J1772 pilot signal circuit, and energy metering IC integration collectively represent knowledge areas that are genuinely rare at the graduate/early-career level.

From a portfolio perspective, this project achieves what employers most value: it demonstrates that the candidate can take a complex, multi-disciplinary engineering problem, decompose it into a structured system architecture, and execute it with engineering rigour. The open-source publication of the firmware, schematics, and documentation on GitHub maximises visibility and verifiability — allowing prospective employers and collaborators to inspect the technical substance of the work directly.

As EV adoption accelerates across Africa and globally, the engineers with deep competency in EV infrastructure, embedded real-time systems, and IoT cloud integration will be among the most sought-after in the industry. This project is designed to be exactly that evidence.

Project Repository Structure		
smart-ev-charger-controller/		
├── firmware/	# STM32H7 C firmware (CMake + STM32CubeIDE)	
├── Core/	# FreeRTOS tasks, state machine, main	
├── Drivers/	# ATM90E36A, MFRC522, SSD1306, ACS758 drivers	
├── Comms/	# MQTT, CAN, Modbus, OCPP stacks	
└── Tests/	# Unity unit test suite	
├── hardware/	# KiCad schematic, PCB layout, BOM	
├── cloud/	# Firebase config, Cloud Functions (Node.js)	
├── dashboard/	# React.js admin dashboard	
├── mobile/	# PWA user application	
└── docs/	# This document + architecture diagrams	